



OBJECT ORIENTED PROGRAMMING

SECTION 02

MINI PROJECT REPORT

TITLE:

Foods Ordering System

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Introduction

With the rapid growth of digital technology, food ordering systems have become increasingly popular, especially among small and medium-sized food businesses such as burgers. The main objective of this project is to develop a simple and functional Food Ordering System using Object-Oriented Programming (OOP) principles in Java.

The Food Ordering System is a software application designed to automate the process of ordering foods. Our system mainly sells baked goods such as breads and muffins. This system also allows customers to browse menus for baked goods with detailed descriptions, place orders, make payments, and track their orders efficiently. It replaces traditional manual ordering methods with a digital platform, improving accuracy, speed, and customer satisfaction.

Problem Statement

A) Current Challenges in Traditional Bakery Order Management

Small bakeries often struggle with manual order-taking processes, which can lead to:

- **Order Confusion:** Without a clear digital system, it's hard to manage orders placed by walk-in customers and recurring ones.
- **Limited User Management:** Lack of secure login/registration makes it difficult to track customer activities.
- **No Role Differentiation:** Bakery systems often don't separate customer and admin privileges, leading to potential security and usability issues.
- **Menu and Price Miscommunication:** Without a standardized system, customers may receive outdated information about products and prices.

- **Lack of Order History and Records:** Manual systems don't store historical order data, making it hard to trace purchases or generate reports.

B) Proposed Solutions

To overcome these challenges, our Food Ordering System introduces:

1) User Authentication and Roles

- **Login and Registration System:** Enables customers to create accounts and securely log in.
- **Role-Based Access:** Distinguishes between Admin and Customer roles, with the Admin managing menu and customers placing orders.

2) Menu and Ordering Features

- **Dynamic Bakery Menu:** The system displays predefined baked items (e.g., Breads, Muffins) with prices.
- **Simple Ordering Process:** Customers can select items from a menu and confirm purchases through a text-based interface.

3) Administrative Functions

- **Menu Management:** Admins can add or remove bakery items.
- **User Overview:** Admins can view all registered users to manage accounts.

4) File-Based Data Storage

- **Persistent User Data:** Stores user credentials and profile information using file handling.
- **Data Parsing:** Information such as roles, phone numbers, and addresses are stored and read using a delimiter (|), ensuring efficient data management.

Objectives

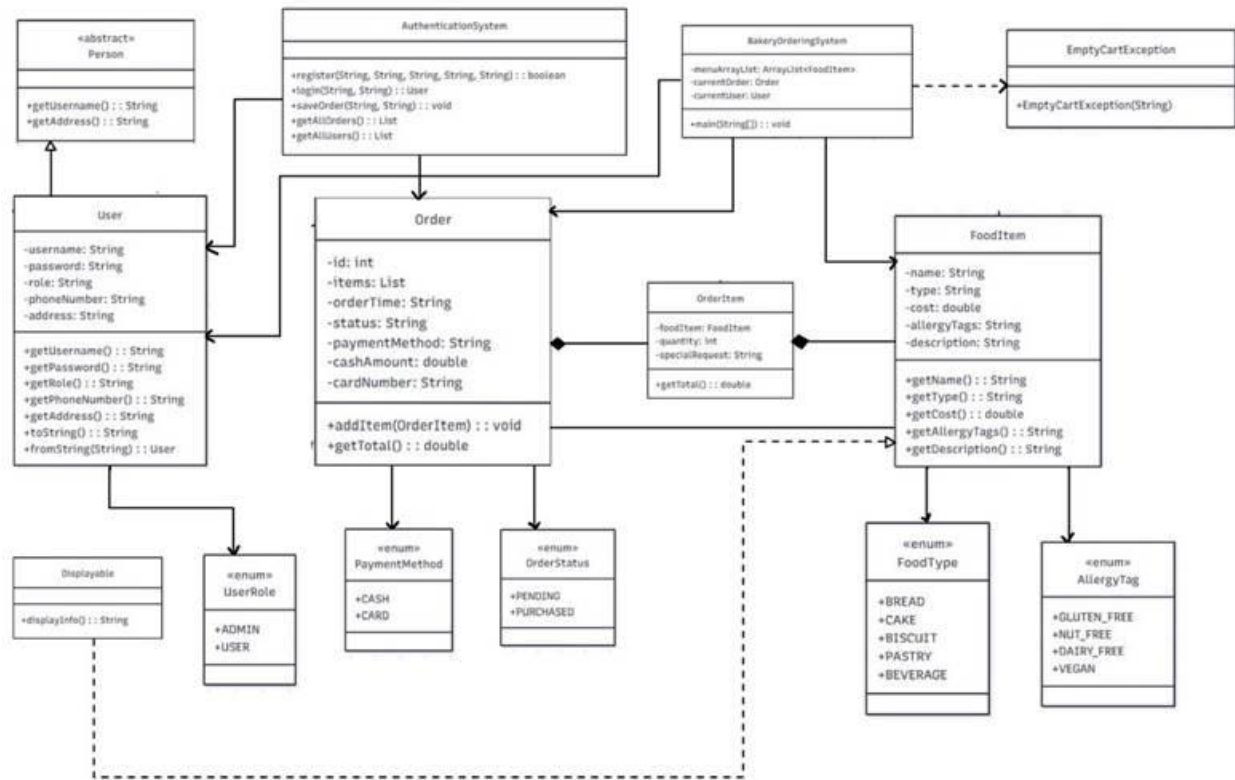
The main objectives of the Food Ordering System project are:

1. To implement object-oriented programming (OOP) principles in Java to develop a modular, reusable, and maintainable food ordering application.
2. To automate the food ordering process, allowing customers to easily browse the bakery menu, place orders, and make payments through a digital interface.
3. To reduce manual errors associated with traditional ordering methods by using structured digital forms and predefined options.
4. To enhance customer convenience and satisfaction through features like real-time order tracking, multiple payment methods, and saved order history.
5. To provide administrative functionality for managing menus, tracking sales, and monitoring inventory levels.

Project Design

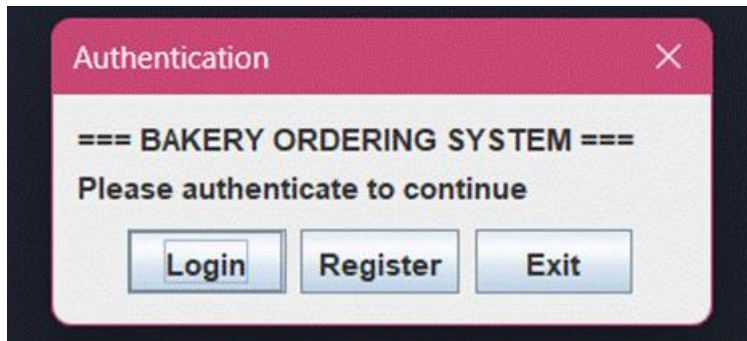
- **Person** is an abstract class that defines the basic structure for users, with **User** inheriting from it and adding attributes such as username, password, role, phone number, and address.
- **FoodItem** represents individual menu items, each with a type (e.g., bread, cake), cost, description, and a set of allergy tags.
- **OrderItem** associates a **FoodItem** with a quantity and any special requests, and multiple **OrderItem** objects are aggregated into an **Order**.
- **Order** manages a collection of order items, tracks the order status, payment method, and relevant payment details.
- **AuthenticationSystem** handles user registration, login, and order management, interacting with both **User** and **Order** classes.
- **BakeryOrderingSystem** is the main class that coordinates user interaction, menu browsing, cart management, and payment processing.
- The system uses several enums (**FoodType**, **AllergyTag**, **OrderStatus**, **PaymentMethod**, **UserRole**) to represent fixed sets of possible values for food types, dietary restrictions, order status, payment methods, and user roles.
- The **Displayable** interface is implemented by **FoodItem** to provide a standardized way to display item information.

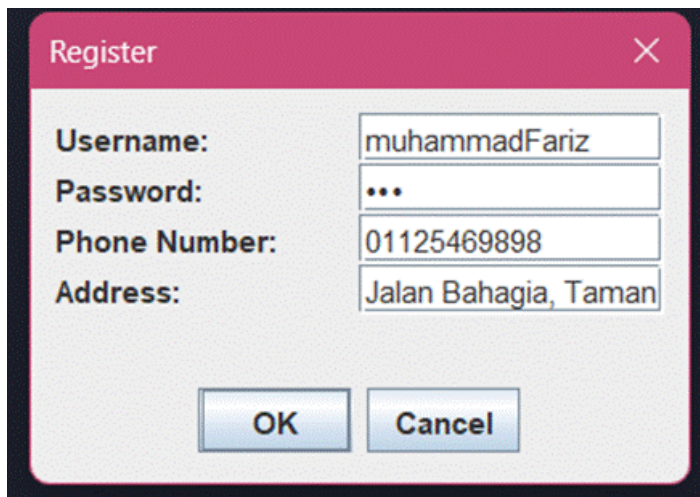
The diagram also shows the relationships between classes, such as inheritance (e.g., **User** inherits from **Person**), interface implementation, and associations (e.g., **Order** contains **OrderItem**, **OrderItem** contains **FoodItem**). Enums are linked to the classes that use them as attributes, clarifying how types and statuses are managed throughout the system.



Project Output

Authentication Feature, when the code is run, the user will have to register into the system first to access the other functions.

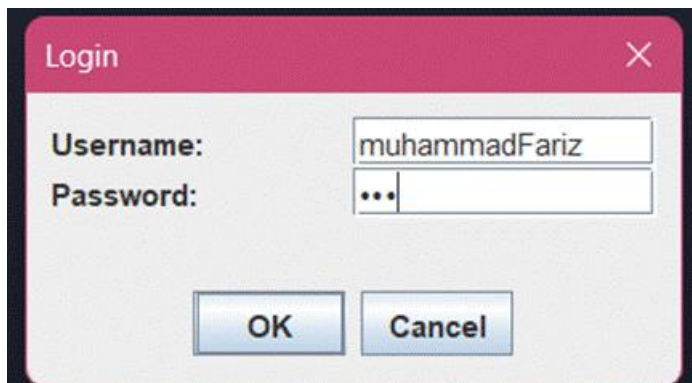




A dialog box titled "Register" with a close button (X) in the top right corner. It contains four input fields: "Username:" with the text "muhammadFariz", "Password:" with three dots, "Phone Number:" with the text "01125469898", and "Address:" with the text "Jalan Bahagia, Taman". At the bottom are two buttons: "OK" and "Cancel".

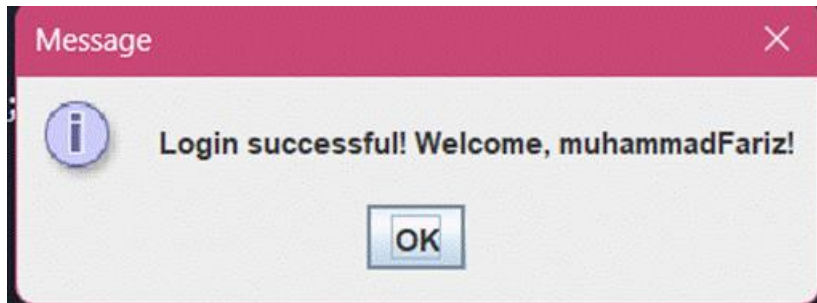
After the user has registered, they will be able to login to the system. Their details will be written in users.txt file.

```
users.txt
1 admin|admin123|ADMIN|0123456789|Admin Address
2 syarifahdania|dania0805|USER|0187732577|johor
3 kokomelon|123|USER|01234654|jaja
4 helloworld|123|USER|0187732577|pulai perdana
5 hehe|123|USER|
6 muhammadFariz|123|USER|01125469898|Jalan Bahagia, Taman Puteri, 50000
7
```



A dialog box titled "Login" with a close button (X) in the top right corner. It contains two input fields: "Username:" with the text "muhammadFariz" and "Password:" with three dots. At the bottom are two buttons: "OK" and "Cancel".

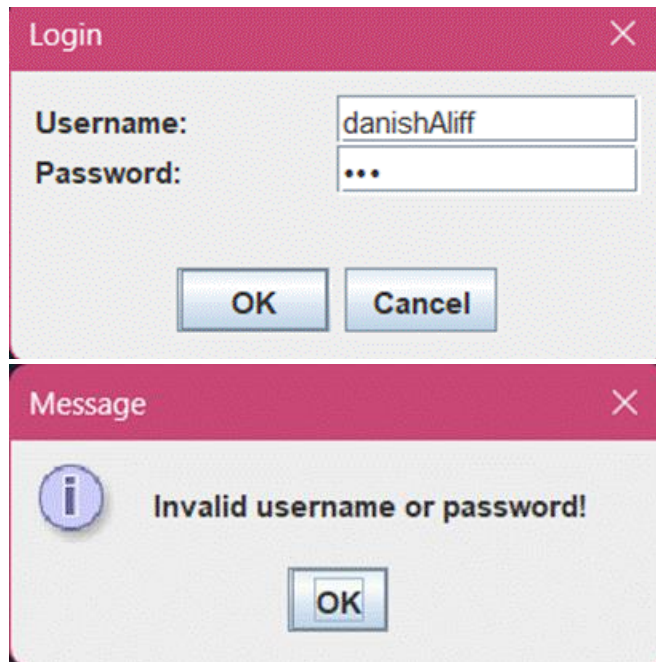
A welcoming message will be shown once the user has successfully logged in.



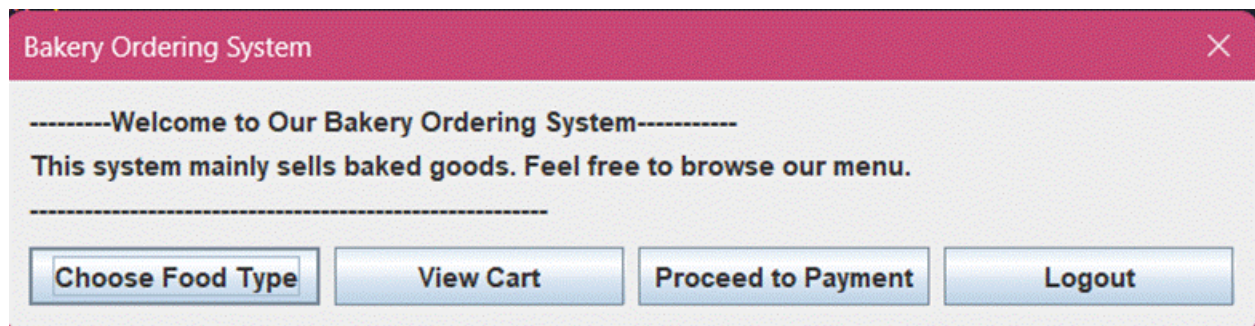
Username is unique for each users. Other users are not allowed to use the usernames that has been registered in the system.

A screenshot of a 'Register' dialog box. The title bar is pink and says 'Register' with a close button (X) on the right. The main area is light gray. It contains four labeled input fields: 'Username:' with the text 'muhammadFariz', 'Password:' with three dots, 'Phone Number:' with the text '0198563214', and 'Address:' with the text 'Bandar Seri Begawan'. At the bottom, there are two blue buttons: 'OK' and 'Cancel'.

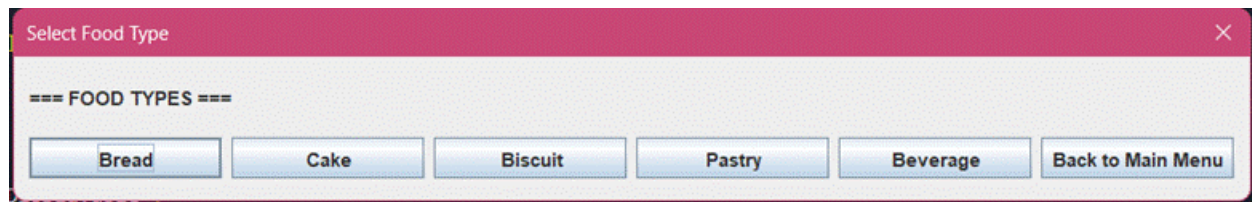
If a user has never registered their account in the system before, or the user keys in wrong password, a warning message will be shown, "Invalid username or password!"



Once the user has logged into our Bakery Ordering System, they will be able to see the main menu of the system which has the function Choose Food Type, View Cart, Proceed to Payment and Logout.

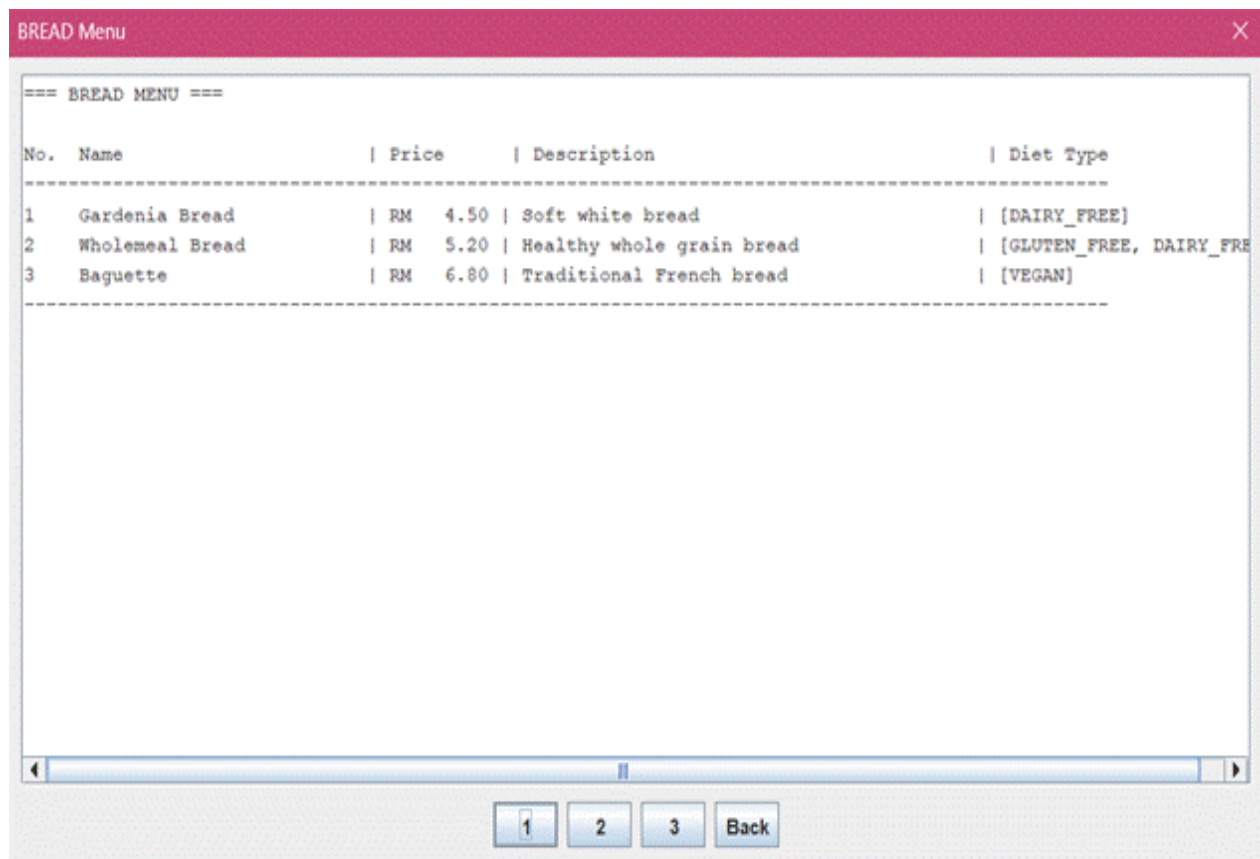


Under Choose Food Type feature, the user will be able to choose a variety of Food Types which are Bread, Cake, Biscuit, Pastry and Beverage. After choosing, the user can press Back to Main Menu to get navigate back to the main menu. Under each food menus, the user can choose what food they want by pressing the number according to the menu.



Menu Based on the Food Types

Bread



Cake

CAKE Menu

×

=== CAKE MENU ===

No.	Name	Price	Description	Diet Type
1	Chocolate Cake	RM 45.00	Rich chocolate flavor	-
2	Red Velvet Cake	RM 55.00	Classic red velvet cake with cream ch...	[NUT_FREE]
3	Cheesecake	RM 38.00	Creamy New York style	[GLUTEN_FREE]

1

2

3

Back

Biscuit

BISCUIT Menu

×

=== BISCUIT MENU ===

No.	Name	Price	Description	Diet Type
1	Butter Cookies	RM 12.50	Buttery and crumbly	-
2	Almond Biscotti	RM 15.00	Italian almond biscuits	[DAIRY_FREE]
3	Shortbread	RM 10.80	Scottish classic	[NUT_FREE]

1

2

3

Back

Pastry

PASTRY Menu

=== PASTRY MENU ===

No.	Name	Price	Description	Diet Type
1	Croissant	RM 6.80	Flaky French pastry	-
2	Danish Pastry	RM 7.50	Sweet fruit-filled pastry	[NUT_FREE]
3	Cinnamon Roll	RM 8.20	Goopy cinnamon goodness	[VEGAN]

1

2

3

Back

Beverages

BEVERAGE Menu

=== BEVERAGE MENU ===

No.	Name	Price	Description	Diet Type
1	Iced Latte	RM 8.50	Cold coffee with milk	[VEGAN]
2	Green Tea	RM 5.00	Refreshing Japanese green tea	[GLUTEN_FREE, VEGAN]
3	Hot Chocolate	RM 7.20	Creamy chocolate drink	[NUT_FREE]

1

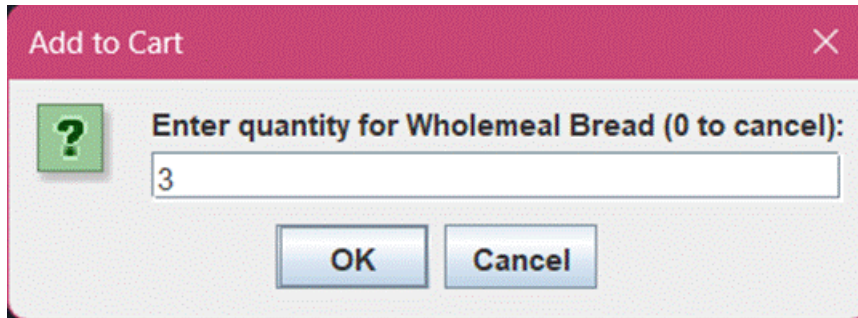
2

3

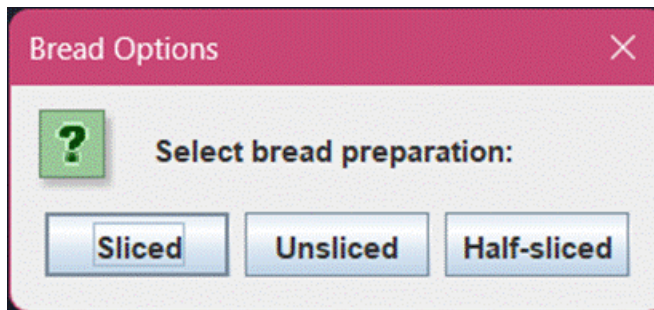
Back

Next, the user can enter the quantity or the amount of food they want. They can also select options or special request for certain food types.

Bread



A dialog box titled "Add to Cart" with a close button (X) in the top right corner. It features a green question mark icon on the left. The main text reads "Enter quantity for Wholemeal Bread (0 to cancel):". Below this text is a text input field containing the number "3". At the bottom, there are two buttons: "OK" and "Cancel".



A dialog box titled "Bread Options" with a close button (X) in the top right corner. It features a green question mark icon on the left. The main text reads "Select bread preparation:". Below this text are three buttons: "Sliced", "Unsliced", and "Half-sliced".



A dialog box titled "Message" with a close button (X) in the top right corner. It features a blue information icon (i) on the left. The main text reads "3 x Wholemeal Bread added to cart!". At the bottom, there is a single button labeled "OK".

Cake

Add to Cart ✕

 Enter quantity for Chocolate Cake (0 to cancel):

Cake Size ✕

 Select cake size:

Message ✕

 1 x Chocolate Cake added to cart!

Biscuit

Add to Cart

?

Enter quantity for Almond Biscotti (0 to cancel):

2

OK

Cancel

Message

i

2 x Almond Biscotti added to cart!

OK

Pastry

Add to Cart

?

Enter quantity for Cinnamon Roll (0 to cancel):

3

OK

Cancel

Message

i

3 x Cinnamon Roll added to cart!

OK

Beverages

Add to Cart

?

Enter quantity for Iced Latte (0 to cancel):

2

OK

Cancel

Beverage Options

?

Select beverage preference:

Regular

Less Ice

Less Sugar

Less Ice & Sugar

No Ice

No Sugar

Message

i

2 x Iced Latte added to cart!

OK

After selecting the foods of preference, the user then can click “Back to Main Menu” to view the cart and make payment.

View Cart

Your Cart

=== ORDER RECEIPT ===
Order Time: 2025-06-27 12:21

No	Name	Type	Special Req	Price	Qty	Subtotal	Description
1	Cheesecake	cake	1 Piece	RM38.00	1	RM38.00	Creamy New York style
2	Wholemeal Bread	bread	Unsliced	RM5.20	3	RM15.60	Healthy whole grain bread
3	Chocolate Cake	cake	1 Piece	RM45.00	1	RM45.00	Rich chocolate flavor
4	Almond Biscotti	biscuit	-	RM15.00	2	RM30.00	Italian almond biscuits
5	Cinnamon Roll	pastry	-	RM8.20	3	RM24.60	Goosey cinnamon goodness
6	Iced Latte	beverage	Less Sugar	RM8.50	2	RM17.00	Cold coffee with milk
7	Baguette	bread	Sliced	RM6.80	2	RM13.60	Traditional French bread

TOTAL : RM183.80

ORDER STATUS: PENDING

Remove Item

Back to Main Menu

In the cart, the user is able to remove item of choice by selecting Remove Item.

Remove Item

?

Enter item number to remove:

7

OK

Cancel

Message

i

Removed: 2 x Baguette

OK

Below is the cart menu after removing the 7th item which is Baguette.

Your Cart

=== ORDER RECEIPT ===
Order Time: 2025-06-27 12:21

No	Name	Type	Special Req	Price	Qty	Subtotal	Description
1	Cheesecake	cake	1 Piece	RM38.00	1	RM38.00	Creamy New York style
2	Wholemeal Bread	bread	Unslice	RM5.20	3	RM15.60	Healthy whole grain bread
3	Chocolate Cake	cake	1 Piece	RM45.00	1	RM45.00	Rich chocolate flavor
4	Almond Biscotti	biscuit	-	RM15.00	2	RM30.00	Italian almond biscuits
5	Cinnamon Roll	pastry	-	RM8.20	3	RM24.60	Gooey cinnamon goodness
6	Iced Latte	beverage	Less Sugar	RM8.50	2	RM17.00	Cold coffee with milk

TOTAL : RM170.20

ORDER STATUS: PENDING

Remove Item

Back to Main Menu

Payment Method

The user will have to choose the payment method. The receipt generated will mention the payment method (Paid by Cash or Paid by Card). Order status will change from PENDING to PAID.

Payment

?

Select a payment method:

Cash

Card

Cash

Cash Payment [X]

Total amount: RM170.20

Enter cash amount:

200

OK **Cancel**

Receipt [X]

i

=== ORDER RECEIPT ===
Order Time: 2025-06-27 12:21

No	Name	Type	Special Req	Price	Qty	Subtotal	Description
1	Cheesecake	cake	1 Piece	RM38.00	1	RM38.00	Creamy New York style
2	Wholemeal Bread	bread	Un sliced	RM5.20	3	RM15.60	Healthy whole grain bread
3	Chocolate Cake	cake	1 Piece	RM45.00	1	RM45.00	Rich chocolate flavor
4	Almond Biscotti	biscuit	-	RM15.00	2	RM30.00	Italian almond biscuits
5	Cinnamon Roll	pastry	-	RM8.20	3	RM24.60	Goey cinnamon goodness
6	Iced Latte	beverage	Less Sugar	RM8.50	2	RM17.00	Cold coffee with milk

TOTAL : RM170.20
CASH : RM200.00
CHANGE : RM29.80
Paid by cash

ORDER STATUS: PAID

OK

Card

Card Payment

×

Enter card number (format: xxxx-xxxx-xxxx-xxxx):

5624-5666-4100-9995

OK

Cancel

Receipt

×

i

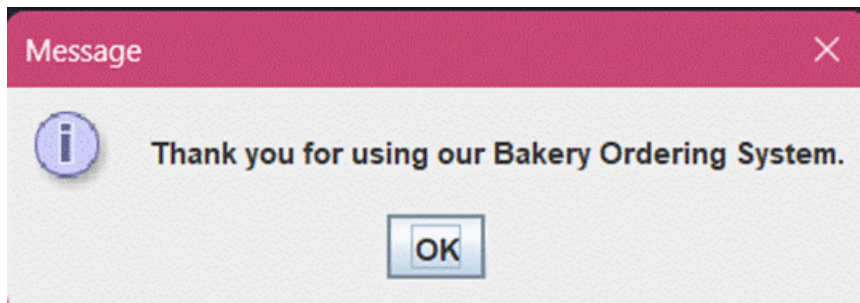
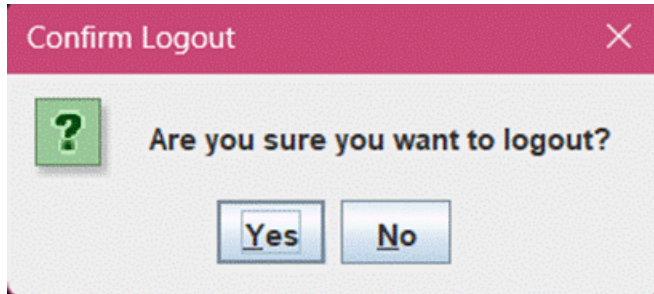
=== ORDER RECEIPT ===
Order Time: 2025-06-27 12:29

No	Name	Type	Special Req	Price	Qty	Subtotal	Description
1	Wholemeal Bread	bread	Sliced	RM5.20	9	RM46.80	Healthy whole grain bread
2	Danish Pastry	pastry	-	RM7.50	4	RM30.00	Sweet fruit-filled pastry
3	Chocolate Cake	cake	1 Piece	RM45.00	2	RM90.00	Rich chocolate flavor

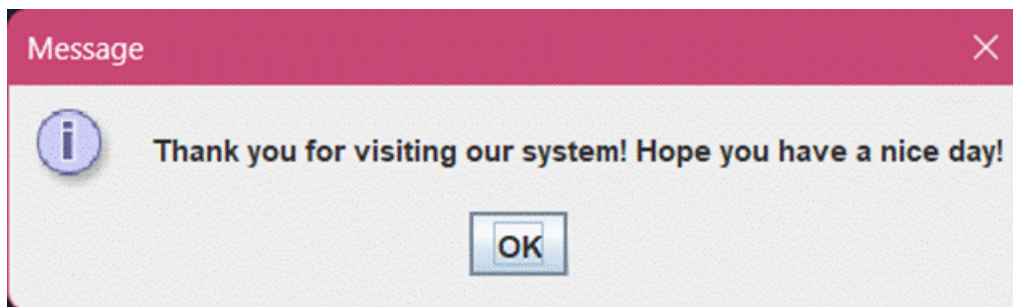
TOTAL : RM166.80
Paid by card
ORDER STATUS: PAID

OK

After successfully making the payment, the user can press OK and they will be navigated back to the main menu. In the main menu, the user can press Logout. They will get a message saying “Are you sure you want to log out?”. If the user presses Yes, they will be brought back to the Authentication menu and they can decide whether to login or register or exit the system. If they press No, the user will be brought back to the main menu.



Below is the output when user presses Exit button on the Authentication menu.



The receipt generated will also be written in a orders.txt file.

```
=== Order by: muhammadFariz ===

=== ORDER RECEIPT ===
Order Time: 2025-06-27 12:21

No  Name                Type      Special Req  Price   Qty   Subtotal  Description          Diet Type
=====
1   Cheesecake           cake      1 Piece      RM38.00  1     RM38.00   Creamy New York style GLUTEN_FREE
2   Wholemeal Bread       bread     Unsliced     RM5.20   3     RM15.60   Healthy whole grain bread GLUTEN_FREE, DAIRY_FREE
3   Chocolate Cake       cake      1 Piece      RM45.00  1     RM45.00   Rich chocolate flavor -
4   Almond Biscotti       biscuit   -            RM15.00  2     RM30.00   Italian almond biscuits DAIRY_FREE
5   Cinnamon Roll         pastry    -            RM8.20   3     RM24.60   Goey cinnamon goodness VEGAN
6   Iced Latte            beverage  Less Sugar    RM8.50   2     RM17.00   Cold coffee with milk  VEGAN
=====

TOTAL                : RM170.20
CASH                  : RM200.00
CHANGE                : RM29.80
Paid by cash

ORDER STATUS: PAID
```

Conclusion

In conclusion, the Food Ordering System successfully demonstrates the application of Object-Oriented Programming concepts in a real-world scenario. The system enhances the efficiency and accuracy of bakery operations by replacing manual ordering with a streamlined digital process. Customers benefit from an intuitive interface, flexible payment options, and real-time order tracking. This project not only meets the objectives outlined but also provides a scalable foundation for future enhancements .

Appendices

The appendices contain the BakeryOrderingSystem.java, users.txt and orders.txt

<https://drive.google.com/drive/folders/1NM7a7oQRRX8lXBIY3TdXl-DXy2HEgRni?usp=sharing>